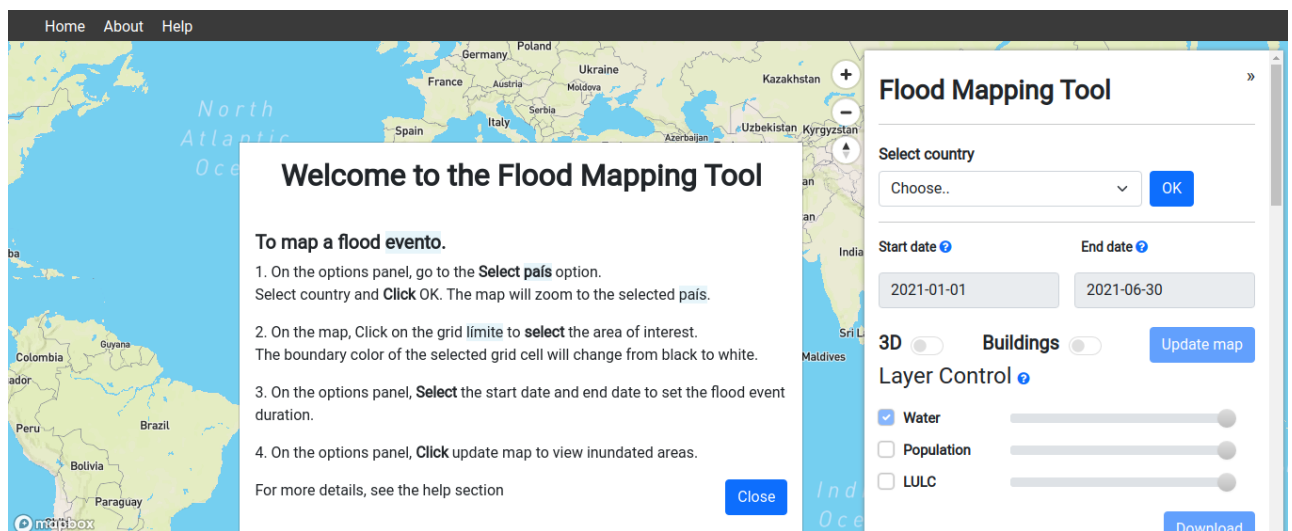


Flood Mapping Tool: Helping Cope with Floods Amidst Climate Change



In times of climate change, software developers across the world have worked together to create an open source online tool to predict floods and understand their impact much better.



The World Flood Mapping Tool aims to “help plan urban and agricultural development, effective flood defences, disaster readiness, and identify supply chain vulnerabilities,” the team that worked on it has said.

“Flooding disasters are a recurring event. Every year, the intensity and frequency of floods is going up. When there’s a flood, there’s a rush of activity. We try to manage that particular event, and then we forget about it. We wait for the next one,” said UNU-INWEH GIS and remote sensing specialist Dr Hamid Mehmood in an exclusive online interview.

But researchers wanting to learn from previous floods can’t map out historical floods. They lack access to

the data. Existing inundation and flood risk maps in most Third World countries are out-of-date and do not consider rapid urban development or the impact of climate change.

A majority of forecasting centres in flood-prone countries lack the ability to improve the spatial coverage of flood early warning systems (FEWS), and generate historical inundation and flood risk maps. Developing these maps nationally with conventional techniques is costly.

That’s where the Flood Mapping Tool (FMT) makes a difference. It promises to produce accurate historical inundation maps. It “bypasses the high-cost issue by using big Earth data, AI models, open data, and cloud computing.” It is expected to provide critical input to flood

mitigation and emergency response, land use planning and investment in resilient infrastructure, insurance schemes, and overall public awareness of flood risks.

Data from 1985

This online tool maps floods from 1985 onwards, right up to the street level. It has been created by the UN University’s Institute for Water, Environment and Health (UNU-INWEH) with the support of Google and MapBox, among others. Based in Hamilton, Ontario, this research and training institute of the United Nations University is a “think tank on water” and has been set up to address water scarcity and access issues internationally. The United Nations University, or UNU, is itself the Tokyo based academic and research arm of the UN.